

YourWalk — v1.1 methodology sign-off

Meeting with Nicole Kalms

Friday 3 July 2026

City of Casey pilot · Monash XYX Lab review

Repo: github.com/AAisenberg/yourwalk

Purpose: Sign off the **scoring methodology v1.1** (how datasets join to footpath segments and become Day/Night Index scores). This unlocks PostGIS load and Q3 app build. **Not in scope today:** routing UI, community submissions, or Council dashboard build.

INDEX MODEL (LOCKED)

Day Index = Footpath Accessibility 60% + Heat & Shade 40% | Night Index = Footpath Accessibility 60% + Lighting / After Dark 40%

Higher score = better walking conditions. ~27,458 T1EAM footpath segments (incl. shared use). Scoring unit: street segment (ADR-008).

How the maths works (plain language)

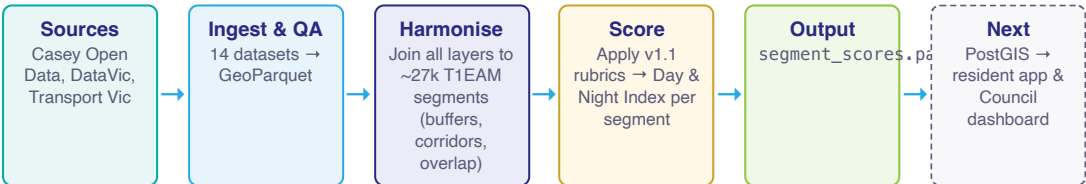
There are two layers of “percent” — don’t mix them up.

- **Layer 1 — the final index:** Accessibility counts for **60%** of Day Index and Night Index. Heat/shade (day) or lighting (night) counts for the other **40%**.
- **Layer 2 — inside Accessibility:** We first build one **Accessibility score from 0 to 100** for each footpath segment. Width, surface, speed, and graffiti each get their own 0–100 sub-score, then blend into that single number.

Why width + surface + speed + graffiti = 30 + 25 + 25 + 15 = 95%? Those weights are relative importance. The code **renormalises** them so they always sum to 100% of the base Accessibility score (divide by 0.95). If speed data is missing (~18% of segments), the other three re-scale automatically.

School crossing “+5” = five points on the 0–100 Accessibility scale — not 5%, not a weight. Added only when a school crossing is within 20 m. No crossing → no penalty. On the final Day or Night Index, +5 Accessibility points ≈ **+3 index points** (because Accessibility is 60% of the index).

A — DATA PIPELINE (ONE ROW PER FOOTPATH SEGMENT)



B — HOW ONE SEGMENT BECOMES TWO INDEX SCORES

- 1 **Harmonise** attaches attributes to each footpath polygon → `segment_features.parquet` (width, surface, speed in 25 m corridor, lights within 30 m, heat overlap, etc.)
- 2 **Score** applies rubrics to those attributes → sub-scores blended into Accessibility, Heat & Shade, and Lighting / After Dark streams

ACCESSIBILITY STREAM → score 0–100			
Width	30%	Surface	25%
Speed exposure	25%	Graffiti proxy	15%
School crossing nearby			+5 pts
Weights renormalise to 100% of base score same Accessibility score × 60% feeds both			
DAY INDEX		NIGHT INDEX	
Heat 2018	45%	Lights (30 m)	70%
Canopy	40%	Night crashes	30%
Comfort	15%	→ After Dark 0–100	
→ Heat & Shade 0–100		0.6 × Acc + 0.4 × After Dark	

Example segment: width 70, surface 90, speed 85, graffiti 100 → base Accessibility ≈ **83**. School crossing nearby → **88** (+5 pts). No school crossing → stays **83**. If Heat & Shade = 75, Day Index ≈ $0.60 \times 88 + 0.40 \times 75 = 83$ (displayed as 8.3/10 in the app).

HARMONISATION → SCORING (BY STREAM)

Dataset	How it joins to each footpath segment	Scoring use	Weight	Notes
Footpaths	Master network — one row per segment	Width + surface (each scored 0–100)	30% 25%	Of Accessibility base (renormalised to 100%). Width: AS 1428-style bands. Surface: 3 buckets + unknown (below). Missing width → neutral 50, not zero.
Speed zones	25 m corridor around footpath; max speed of any road in corridor (parallel arterials, not just crossings)	Speed exposure (0–100)	25%	Slower road beside path = higher score. Missing speed → dropped; other weights re-scale.
Graffiti	Count within 25 m ; recency + recent tags	Maintenance proxy (0–100)	15%	Same Accessibility score feeds both Day and Night Index. Not crime data.
School crossings	Nearest crossing ≤ 20 m from footpath	+5 points (not %)	after blend	142 school crossings only. +5 pts on 0–100 Accessibility if nearby; +0 if not. No school-hours logic . Does not replace general crossing data.
Urban heat 2018	Area-weighted mesh-block overlap → <code>uhi18_m</code>	Heat sub-score	45% of Heat&Shade	Heat&Shade is 40% of Day Index only . Lower UHI = better. 2018 vintage documented.
Tree density	Area-weighted dense / medium / sparse polygons	Canopy sub-score (0–100)	40% of Heat&Shade	2019/2020 Vicmap. Council tree points not used for canopy.
Fountains + benches	Nearest fountain; bench count in 50 m	Comfort sub-score (0–100)	15% of Heat&Shade	Presence only (bench quantity fields unreliable).
Street + park lights	Nearest distance; count within 30 m (street) / 50 m (park); merged	Lighting sub-score (0–100)	70% of Night stream	Night stream = 40% of Night Index . Good if nearest ≤25 m and ≥1 light in 30 m. No lux — proximity only.
Night crashes	Pedestrian crashes in 25 m corridor; night-eligible; 5-year window; density per km	Crash sub-score (0–100)	30% of Night stream	Night Index only (not combined day/night crash scoring).

Surface buckets (Accessibility)

- **Smooth / firm** → **90** — concrete, asphalt, spray seal
- **Moderate** → **50** — brick paving, crushed rock, timber (Nikki 3 Jul 2026)
- **Rough / loose** → **35** — gravel, rubber
- **Unknown** → **50** — TBD, NULL

Explicit gaps — v1.1 deferred

- **General crossings + kerb ramps** — not on open data; reduced confidence, not zero penalty
- **Gradient** — deferred v1.2 — `docs/GRADIENT_DISCOVERY.md`
- **Traffic signal phasing** — deferred v1.2+ — `docs/TRAFFIC_SIGNAL_PHASING.md`
- **YourGround, toilets, dog bags** — dashboard overlays only, not in index

DEMO & REPO REFERENCES

Scoring map: `docs/meeting-prep/casey-scoring-map.html` · Specs: `docs/VULNERABILITY_INDEX.md`, `docs/SEGMENT_HARMONISATION.md`, `docs/SCORING_SPEC_v1.1.md` · Gradient discovery: `docs/GRADIENT_DISCOVERY.md` · Signal phasing: `docs/TRAFFIC_SIGNAL_PHASING.md` · Sign-off log: `docs/meeting-prep/NIKKI_SIGNOFF_DECISIONS.md`

DECISIONS FOR SIGN-OFF TODAY

- ☐ Accept Day/Night 60/40 index model and segment-level scoring as implemented
- ☐ Confirm speed corridor (25 m), lighting (30 m count / 25 m nearest), crash (25 m / 5 yr) join rules
- ☐ Confirm missing crossings / kerb ramps / gradient handled as **gap + reduced confidence**, not imputed zeros or optimistic baselines

- ☐ Confirm school crossings: **+5 points** on Accessibility (not 5%); static bonus only — no school-hours weighting in v1.1
 - ☐ Moderate surface bucket → **50** (brick paving, crushed rock, timber) — Nikki 3 Jul 2026 (v1.1.2 patch)
 - ☐ Accept 2018 heat vintage with documented limitation for pilot
 - ☐ Minor rubric tweaks after today → v1.1.x patch, not a methodology reopen
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Sign-off — v1.1 scoring methodology accepted for City of Casey pilot:

Name: _____ Role: _____ Date: _____

Notes / rubric adjustments:

YourWalk · CrowdLab / Monash XYX Lab · Methodology source of truth in repo (Notion mirrors status only). After sign-off: PostGIS load → resolve app ADRs → Q3 MVP build.